पैदावार™
टिकाऊ कृषि के साथ

Modern Farming Solutions Handbook

"Your Farming, Our Responsibility"

Seasonal Crop Protection at a Glance

■ KHARIF
June – October

- Rice, Maize, Cotton
- Soybean, Groundnut
- Arhar, Bajra
- Drain fields before flood

RABI
October – March

- Wheat, Barley, Gram
- Mustard, Lentil
- Potato, Pea
- Irrigate at flowering

■ ZAID
March – June

- Cucumber, Watermelon
- Moong, Cowpea
- Sunflower, Okra
- Mulch to save moisture

Threat	Kharif Response	Rabi Response	Zaid Response
Excess Rain	Open drainage channels; sow flood-tolerant varieties (Swarna Sub-1)	Avoid sowing on waterlogged soil; add raised beds	Light irrigation only; stop if raining
Drought	Apply mulch; use drip irrigation; grow short-duration varieties	Deficit irrigation at critical stages; potassium spray	Irrigate every 5 days; use plastic mulch
Heat Stress	Spray kaolin 5%; irrigate at midday heat; avoid noon sowing	Harvest early morning; avoid delay beyond maturity	Shade nets 50%; sow in evening; frequent irrigation
Frost / Cold	Not typical for Kharif; protect nurseries with polythene	Light irrigation before frost night; smoke technique in orchards	Use row covers; apply K-rich fertilizer in Oct
Pest Surge	Deploy pheromone traps; spray Bt or neem at ETL	Yellow rust: Propiconazole 1ml/L; aphid: neem oil 3ml/L	Yellow sticky traps; garlic-chilli spray weekly

12 Chapters	3 Seasons	50+ Key Topics	100+ Tips	FREE Helplines

Edition 2026 | For Farmers of India | Free Distribution

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1 Season-Based Crop Guidance

Kharif · Rabi · Zaid

Kharif Season (June – October)

- **Kharif Crops: Sown during June–July with monsoon; ensure proper drainage and seed spacing.**
- Major crops: Rice, Maize, Cotton, Groundnut, Soybean, Arhar, Bajra, Jowar.
- Ensure field bunding and drainage channels before sowing to prevent waterlogging.
- **Use certified high-yielding varieties (HYV) suited to your district's rainfall pattern.**
- Apply basal NPK dose at sowing; monitor for stem borer in rice and bollworm in cotton.
- Flood-tolerant varieties: Swarna Sub-1 and Sambha Mahsuri Sub-1 in flood-prone areas.

★ **Key Point: Select crops based on climate and sowing window for maximum yield.**

Kharif Tip

Apply 10–15tonnes FYM/ha before monsoon onset to boost organic matter. Sow within 7 days of first reliable 50mm rain for best germination.

Rabi Season (October – March)

- **Rabi Crops: Sown in Oct–Nov; require timely irrigation and cooler climate.**
- Major crops: Wheat, Barley, Mustard, Gram (Chickpea), Lentil, Peas, Potato.
- **Sow wheat between Oct 15 – Nov 15 for maximum yield; delay reduces yield 30 kg/ha/day.**
- Provide 4–6 irrigations at crown root initiation (21 DAS), tillering, jointing, boot, flowering.
- Watch for yellow rust (Puccinia striiformis) — apply Propiconazole 25EC @1ml/L at first sign.
- **Prepare soil deeply (25–30 cm) — good tilth ensures uniform germination and root establishment.**

★ **Key Point: Timely sowing in Rabi is the single biggest factor for high wheat and mustard yield.**

Zaid / Summer Season (March – June)

- **Zaid Crops: Summer crops grown with irrigation between Kharif and Rabi seasons.**
- Major crops: Cucumber, Watermelon, Muskmelon, Moong, Cowpea, Sunflower, Okra.
- Irrigate every 5–7 days due to high evapotranspiration in summer heat (40°C+).
- **Mulching with paddy straw or plastic mulch reduces soil moisture loss by up to 40%.**
- Avoid midday sowing — sow seeds in early morning or evening for better germination.

★ **Key Point: Select crops based on climate and available irrigation for Zaid to maximize income.**

2 Soil Testing & Management

pH · Fertility · Soil Health Cards

Why Soil Testing Matters

- **Test your soil at least once every 3 years — more often if yields are declining.**
- A soil test reveals pH, Organic Carbon (OC), N, P, K, S, Zn, Fe, Mn, Cu, Boron.
- **Soil Health Cards (SHC) from government labs are FREE — collect before every season.**
- Optimal crop growth occurs at pH 6.0–7.5; acidic or alkaline soils need amendment.
- **pH Correction: Apply lime (acidic soil) or gypsum (alkaline/sodic soil) as per SHC.**
- Fertility: Improve using compost, FYM 10–15 t/ha, green manure, and crop rotation.

★ **Key Point: Healthy soil directly impacts productivity and profit — never skip soil testing.**

Sample Tip

Collect 10–15sub-samples from 0–15 cm depth in zigzag pattern; mix, air-dry, send 500g to nearest KVK. Results come in 10–15 days free of cost.

Improving Soil Fertility (NEW — Enhanced)

- **Apply vermicompost 2–3 t/ha — significantly improves soil microbes and nutrient availability.**
- Biofertilizers: Rhizobium (legumes), Azospirillum (cereals), PSB (phosphate solubilizing bacteria).
- **Bihar/UP soils are frequently deficient in Zinc and Boron — always include in soil health plan.**
- Zinc deficiency (khaira in rice): Apply $ZnSO_4$ @25 kg/ha or 0.5% foliar spray.
- Sulphur deficiency common in sandy soils — apply gypsum 200–250 kg/ha at sowing.

★ **Key Point: 1 tonne of compost can replace 8–10 kg Urea, 5 kg DAP, and 7 kg MOP — saves input cost.**

3 Climate Protection

Heat · Frost · Flood · Drought

Heat Protection

- **Heat Protection: Apply light irrigation (sprinkling) during heat waves to cool crop canopy.**
- Use kaolin clay spray 5% on leaves to reflect sunlight and reduce leaf temperature by 3–5°C.
- **Choose heat-tolerant varieties: HD-3226 (wheat), DRRH-3 (rice) for high-temperature zones.**
- Provide shade nets (50% shading) for vegetable nurseries during April–May.
- Temperatures above 35°C during wheat flowering can reduce grain weight by 20–30%.

★ **Key Point: Use mulching, shade nets, and irrigation to protect crops from heat stress.**

Frost Protection

- **Frost Protection: Light irrigation before a frost night raises field temperature by 2–3°C.**
- Apply potassium-rich fertilizers (MOP/SOP) in October to improve plant frost resistance.
- Use polythene tunnels or row covers for vegetables (tomato, brinjal) in December–January.
- Smoke/smudge fires upwind of orchards on frost nights protect flowers (traditional method).
- **Prune frost-damaged tissue only after danger passes — wait 2–3 weeks to assess true damage.**

★ **Key Point: Climate adaptation minimizes risk and loss — prepare before the season, not after.**

Flood Management

- **Flood Management: Drain stagnant water within 24–48 hours — beyond this causes root death.**
- Raise seedbed height by 15–20 cm in flood-prone areas before sowing season.
- After floods recede, spray 2% urea solution on leaves for quick nitrogen recovery.
- **Sow flood-tolerant rice varieties: Swarna Sub-1, Sambha Mahsuri Sub-1 in flood zones.**
- Repair bunds immediately after flood water recedes to prevent further soil erosion.

★ **Key Point: Quick drainage and resistant crop varieties minimize flood damage.**

Drought Management

- **Drought: Apply mulch (straw/plastic) to conserve moisture — reduces water loss 30–50%.**
- Use drought-tolerant varieties: GW-496 (wheat), DRR Dhan 42 (rice), Phule Mauli (soybean).
- **Use drip irrigation and drought-tolerant seeds in water-scarce regions.**
- Foliar spray of 1% KNO3 or 1% KCl reduces drought stress and improves plant survival.
- Adopt contour bunding and farm ponds to harvest rainwater for dry period irrigation.

★ **Key Point: Use drip irrigation and drought-tolerant seeds — the two pillars of drought management.**

Water Saving Tip

Deficitirrigation (60–70% of full requirement) at non-critical stages saves 30% water with only 10–15% yield loss — far better than complete crop failure.

4 Irrigation & Water Management

Drip · Sprinkler · Critical Stages

Critical Irrigation Stages

- **Critical Stages: Flowering and grain filling stages are the most sensitive to water stress.**
- Wheat: 5 irrigations at CRI (21 DAS), tillering, jointing, boot, and grain filling stages.
- Rice: Maintain 2–5 cm standing water during tillering; drain 15 days before harvest.
- Maize: Knee-high, tasseling, and grain filling are critical — never miss these timings.
- Mustard: One irrigation at branching-flowering is sufficient for good yield.

★ **Key Point: Efficient water use at critical stages improves yield and sustainability significantly.**

Micro-Irrigation Methods

- **Micro Irrigation: Drip and sprinkler systems save 40–60% water over flood irrigation.**
- Drip irrigation: Best for vegetables, orchards, sugarcane — 90% water use efficiency.
- Sprinkler irrigation: Ideal for wheat, groundnut, vegetables — 75% water efficiency.
- **Subsidies: PM Krishi Sinchayee Yojana provides 55% subsidy on drip/sprinkler systems.**
- Alternate Furrow Irrigation (AFI) for maize/soybean saves 30–40% water.

★ **Key Point: Government schemes support irrigation systems — apply at your nearest agriculture office.**

▪ **Drip Subsidy**

PM KrishiSinchayee Yojana: Apply at agriculture.gov.in or visit District Agriculture Office with land records, Aadhaar card, and bank passbook for 55% subsidy.

5 Fertilizer Management

NPK · Micronutrients · Organic Integration

NPK Split Application

- **NPK Split Application: Apply nitrogen in multiple doses — never all at once.**
- Urea (46% N): Apply in 3 splits — basal at sowing, 30 DAS, 60 DAS for best efficiency.
- DAP (18% N + 46% P): Apply full dose at sowing as basal — P doesn't need splitting.
- **Use neem-coated urea — reduces N losses by 10–15% and improves crop uptake.**
- Potassium (MOP): Apply at sowing or first irrigation — improves disease resistance.

★ **Key Point: Balanced nutrition ensures healthy crop growth — follow Soil Health Card recommendations.**

Micronutrients & Organic Integration

- **Micronutrients: Zinc, Boron, Sulfur are essential and often deficient in Indian soils.**
- Zinc ($ZnSO_4$ @25 kg/ha) for rice, wheat, maize — apply to soil or 0.5% foliar spray.
- Boron (Borax @1 kg/ha) for cauliflower, mustard, cotton — prevents hollow stem.
- **Organic Integration: Combine organic (FYM, vermicompost) and chemical fertilizers.**
- Adding 5 t/ha FYM reduces chemical fertilizer need by 25–30% and improves soil health.

★ **Key Point: Balanced nutrition from both organic and inorganic sources gives the best long-term results.**

6 Pest & Disease Control (IPM)

Bio-control · ETL · Natural Sprays

Integrated Pest Management Principles

- **Biological Control: Use Trichogramma cards, Bt, and neem oil — safe for pollinators.**
- Neem-based pesticides (Azadirachtin 0.03%): Effective against sucking pests, leaf miners.
- **Chemical Use: Apply pesticide ONLY when pest population exceeds Economic Threshold Level.**
- Natural Sprays: Blend 100g garlic + 50g chilli + 1L water; dilute 1:20 — spray weekly.
- Pheromone traps: Use for monitoring Spodoptera, Helicoverpa, Diamond-back moth.
- **Never spray pesticides during flowering — saves bees critical for fruit and grain setting.**

★ **Key Point: Early detection reduces crop damage — scout fields at least twice weekly during crop season.**

Safety First

Always wear gloves, mask, and protective clothing while spraying. Wash hands thoroughly after handling. Store chemicals locked away from children and food items.

Key Disease Management

- **Rice Blast: Spray Tricyclazole 75WP @0.6g/L at neck emergence — critical timing.**
- Wheat Yellow Rust: Apply Propiconazole 25EC @1ml/L at first symptom appearance.
- Early Blight of Tomato: Spray Mancozeb 75WP @2.5g/L every 10 days after infection.
- **Bacterial Leaf Blight (rice): No chemical cure — use resistant varieties; avoid excess N.**
- Powdery Mildew (cucurbits): Apply Sulphur 80WP @3g/L or Karathane 0.5ml/L.

★ **Key Point: Correct identification of disease before spraying saves cost and prevents resistance.**

7 Seed Selection & Treatment

Certified · Treated · Tested

Choosing the Right Seed

- **Certified Seeds: Ensure high germination rate — minimum 85% germination for field sowing.**
- Buy seeds only from NSC, IARI, KVK, State Seed Corporation, or licensed dealers.
- Hybrid seeds give 20–30% higher yield but cannot be saved for next season replanting.
- **Open-pollinated varieties (OPV): Seeds can be saved and reused for 3–4 seasons.**
- Select variety based on district's agro-climatic zone — check local KVK recommendations.

★ **Key Point: Good seed quality determines crop success — never compromise on seed certification.**

Seed Treatment (Before Sowing)

- **Seed Treatment: Treat with Thiram + Carbendazim (2g+1g per kg) before sowing.**
- Rhizobium seed inoculant for legumes: Mix 25g per kg seed — never mix with chemical fungicide.
- Hot water treatment at 52°C for 30 minutes kills seed-borne pathogens in wheat and rice.
- **Always treat seeds in shade — never in direct sunlight as it kills beneficial microbes.**

★ **Key Point: Use fungicides and biofertilizers together (on separate days) for best seedling protection.**

Germination Test

Before large-scale sowing: test 100 seeds on wet cloth for 7 days. Count germinated seeds — if below 80%, increase seed rate by 20% or change the seed batch entirely.

8 Harvesting & Post-Harvest Management

Maturity · Storage · Loss Prevention

Harvest Maturity Indicators

- **Maturity Indicators: Color change and grain hardness are the primary signs of maturity.**
- Wheat: Harvest at 14–18% grain moisture; golden yellow color, 'click' sound on biting.
- Rice: 80–85% grain maturity — yellow panicles, 20–24% grain moisture content.
- Vegetables: Harvest in early morning when turgid — extends shelf life by 30–40%.
- **Delayed harvesting causes shattering losses in wheat/mustard and quality deterioration.**

★ **Key Point: Timely harvesting at correct maturity directly increases farmer income per acre.**

Storage & Post-Harvest Care

- **Storage: Use airtight (hermetic) storage bags — eliminates insect infestation without chemicals.**
- Dry grains to 12–14% moisture before storage — excess moisture causes fungal spoilage.
- **Loss Prevention: Reduce 10–20% losses with proper storage — this is free income recovery!**
- Apply Aluminium Phosphide tablets (3g/tonne) in sealed storage for long-term grain storage.
- Never store seed grain with food grain — keep labelled, dry, and off the ground on pallets.

★ **Key Point: Post-harvest care increases farmer income — proper storage can save Rs.500–1000 per quintal.**

Storage Tip

APICStriple-layer hermetic bag (Rs.60–80) protects 100 kg grain from weevil damage for 6–9 months — pays for itself 10× over. Available at KVK and agri-input shops.

9 Organic Farming

Vermicompost · Jeevamrut · PGS Certification

Building Organic Inputs

- **Vermicompost: Enhances soil fertility — 1 kg earthworms converts 3–4 kg waste daily.**
- **Jeevamrut & Panchagavya: Natural growth boosters that improve crop immunity and yield.**
- Jeevamrut: 10L cow urine + 10 kg dung + 2 kg jaggery + 2 kg pulse flour in 200L water; ferment 48 hrs.
- Green manuring: Bury dhaincha/sunhemp at 50% flowering — adds 80–100 kg N/ha to soil.
- Compost C:N ratio of 20:1 is ideal — mix brown (carbon) and green (nitrogen) materials.
- **PGS Certification: Enables organic product selling at premium price — apply via FPO/APEDA.**

★ **Key Point: Organic farming supports long-term sustainability and can increase income by 20–30%.**

PGS Tip

Participatory Guarantee System (PGS) organic certification takes 3 years of transition. Form a group of 5+ farmers for group certification to reduce cost and effort significantly.

10 Government Schemes & Support

PM-KISAN · PMFBY · KCC · Helplines

Key Schemes Every Farmer Must Know

- **PM-KISAN: Direct financial assistance of Rs.6,000/year in 3 installments — register now!**
- **PMFBY: Crop insurance protection — 1.5–2% premium for Rabi, 2% premium for Kharif crops.**
- KCC (Kisan Credit Card): Easy agricultural loans at 4% effective rate up to Rs.3 lakh.
- Soil Health Card Scheme: FREE soil testing every 3 years — collect at agriculture office.
- PM Krishi Sinchayee Yojana: 55% subsidy on drip/sprinkler irrigation systems.
- Helplines: Free expert advisory — Kisan Call Centre: 1800-180-1551 (Toll Free, 24×7).

★ **Key Point: Use government schemes to reduce financial burden and protect against crop losses.**

Helplines

KisanCall Centre: 1800-180-1551 | PM-KISAN: 155261 | eNAM: 1800-270-0224 | PMFBY: 14447 — All calls are free and available in local languages including Hindi, Bhojpuri.

11 Climate-Smart Agriculture

Zero-Till · SRI · Agroforestry · Residue Mgmt

NEW

Smart Practices for a Changing Climate

- **Zero Tillage: Conserves soil moisture — reduces fuel cost 60–70% and protects soil structure.**
- **SRI Method (System of Rice Intensification): Boosts rice productivity 25–40% with 30% less water.**
- Agroforestry: Trees + crops integration — Poplar/Eucalyptus boundaries reduce wind damage.
- **Residue Management: Avoid stubble burning — use Happy Seeder or incorporate as mulch/compost.**
- Diversify crops — grow at least 3 different crops on farm to reduce total income risk.
- Carbon farming: Zero-till + cover crops + compost earn carbon credits — future income source.

★ **Key Point: Smart farming practices improve resilience and yield while cutting input costs by 20–30%.**

Climate Tip

Zero-till wheatafter paddy saves 6–8 litres diesel/acre, retains soil moisture, and advances sowing by 10–15 days — giving higher yield than conventional tillage wheat.

12 Emergency Crop Recovery Plan

Flood · Drought · Hail · Insurance Claims

NEW

Immediate Actions After Crop Damage

- **Report crop loss to Patwari/Agriculture Department within 72 hours for insurance claim.**
- **Flood: Drain water within 48 hrs, reapply nutrients with 2% urea foliar spray after draining.**
- **Drought: Irrigate immediately, mulch soil surface, consider replanting short-duration moong/cowpea.**
- **Hail Damage: Apply protective copper fungicide spray within 24 hours to prevent infection.**
- Insurance Claims: Report quickly under PMFBY — photograph all damage with date/GPS stamps.
- Contact District Agriculture Officer, BDO, or KVK for emergency seeds and fast-track support.

★ **Key Point: Quick action after disaster reduces major crop losses — act within 48 hours, not 48 days.**

Emergency Contacts

Kisan CallCentre: 1800-180-1551 | State Agriculture Department | District Agriculture Officer (DAO) | Nearest KVK (Krishi Vigyan Kendra) — all can provide emergency guidance.

13 Digital Tools for Farmers

Apps · Mandi Prices · Weather Alerts

NEW

Essential Apps & Online Resources

- **Kisan Suvidha App: Weather forecast, market prices, crop advisories — FREE on Android.**
- eNAM (National Agriculture Market): Sell crops online at best price — register at enam.gov.in.
- **Crop Insurance App (PMFBY): File claims, check status, and see premium calculator.**
- Agrimarket App: Real-time mandi prices from 3,000+ markets across India.
- mKisan Portal: Receive SMS alerts for weather, pest warnings, and crop advisories in Hindi.
- **PM-KISAN App: Check payment status, update Aadhaar, and apply for KCC from phone.**
- IFFCO Kisan: WhatsApp-based advisory in local language — send 'Hi' to +91-99-9009-8989.

★ **Key Point: Digital tools give real-time information to make smarter, faster farming decisions.**

Digital Tip

Even abasicsmartphone can access mKisan weather SMS alerts for FREE. Call 1800-180-1551 to register your number for free crop advisory SMS in your local language.

14 Crop Calendar Quick Reference

Month-wise Action Planner for Farmers

NEW

Month	Season	Key Action	Do NOT
June	Kharif Start	Sow rice/maize; apply FYM; open drainage channels	**Delay sowing beyond Jun 20**
July	Kharif	Top-dress urea; monitor stem borer; weed control	**Leave stagnant water >48 hrs**
Aug	Kharif	2nd urea dose; spray for leaf folder; soil moisture check	**Over-irrigate during monsoon**
Sep	Kharif End	Prepare for harvesting; repair thresher; check storage	**Harvest before physiological maturity**
Oct	Rabi Start	Sow wheat/mustard; apply K fertilizer; soil test	**Delay wheat sowing beyond Nov 15**
Nov	Rabi	CRI irrigation for wheat at 21 DAS; weed removal	**Skip CRI irrigation — most critical stage**
Dec	Rabi	Frost protection; protect mustard from aphid; check pH	**Irrigate before frost night (do it the evening before)**
Jan	Rabi	Jointing irrigation wheat; monitor yellow rust; sprinkle	**Ignore rust spots — spray within 48 hours of detection**
Feb	Rabi	Flowering irrigation; apply urea 3rd dose to wheat	**Over-irrigate at grain filling — reduces protein content**
Mar	Rabi Harvest/Zaid	Harvest mustard/gram; sow moong/cucumber; mulch fields	**Burn crop residue — it destroys soil microbes**
Apr	Zaid	Irrigate every 5 days; apply K; watch for mites in heat	**Spray pesticides at noon — high evaporation, low efficacy**
May	Zaid End/Preparation	Harvest summer crops; deep plough fields; make compost	**Leave fields bare — use green manure/cover crop**

Disclaimer: This handbook provides general educational guidance for farmers across India. Crop recommendations vary by region, soil type, and local climate. Always consult your nearest **Krishi Vigyan Kendra (KVK)**, District Agriculture Officer, or ICAR Research Station for location-specific advice.

Key Helplines: Kisan Call Centre: 1800-180-1551 (Toll Free, 24x7) | PM-KISAN: 155261 | PMFBY: 14447 | eNAM: 1800-270-0224
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